Lab Experiment-13

Aim

To write a Python program to generate a parse tree for a given sentence using a Context-Free Grammar (CFG).

Algorithm

1. Start the program.
2. Import the required classes from the `nltk` library.
3. Define a Context-Free Grammar (CFG) with production rules.
4. Get a sentence as input from the user.
5. Split the sentence into words.
6. Parse the sentence using the defined grammar.
7. Display the parse tree.
8. Stop the program.

Python Code

```
import nltk

from nltk import CFG

grammar = CFG.fromstring("""

S -> NP VP

NP -> Det N

VP -> V NP

Det -> 'the' | 'a'

N -> 'cat' | 'dog' | 'mouse'

V -> 'chased' | 'saw'
```

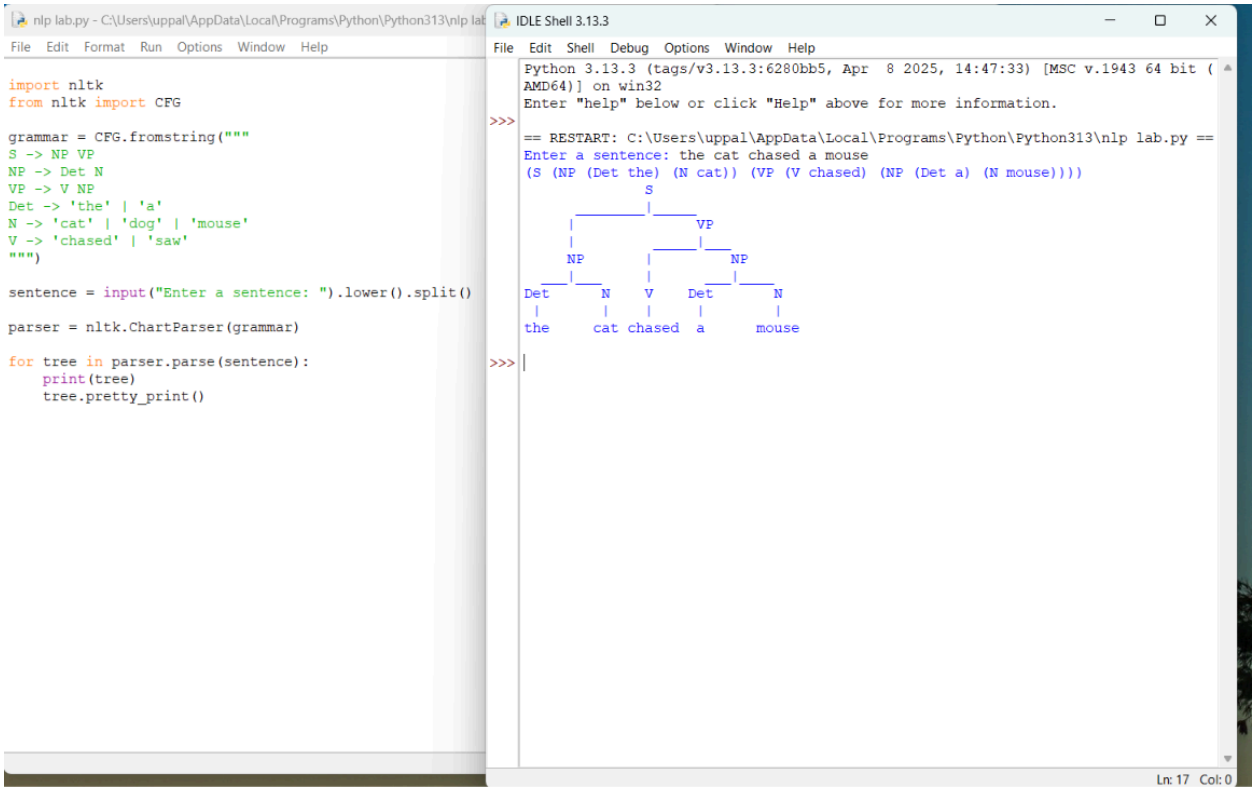
```
""")

sentence = input("Enter a sentence: ").lower().split()

parser = nltk.ChartParser(grammar)

for tree in parser.parse(sentence):
    print(tree)
    tree.pretty_print()
```

Output:



Result: The Python program successfully generates and displays the parse tree for the given sentence using a Context-Free Grammar.